Part 6 — Linux Deep Dive

Probability: 🔴 90% — Very Frequently Asked

100 Linux Interview Questions

File System & Navigation (1-15)

#	Question	Answer / Command
1	Find all log files modified in last 24 hours	`find /var/log -name "*.log" -mtime -1`
2	Find files larger than 100MB	`find / -type f -size +100M 2>/dev/null`
3	Find and delete empty directories	`find /path -type d -empty -delete`
4	Count files in a directory recursively	`find /path -type f | wc -l`
5	What's the difference between hard and soft links?	Hard link: same inode, can't cross filesystems. Soft link: pointer to path, can cross FS, can link dirs
6	Create a symbolic link	`ln -s /target/file /link/name`
7	What is an inode?	Metadata structure storing file info (permissions, owner, timestamps, block pointers). NOT the filename
8	Difference between `/tmp` and `/var/tmp`?	`/tmp` cleared on reboot. `/var/tmp` persists across reboots
9	What is `/proc`?	Virtual filesystem exposing kernel/process info. `/proc/cpuinfo`, `/proc/meminfo`
10	What is `/dev/null`?	Bit bucket — discards all data written to it. `command 2>/dev/null`
11	Explain the Linux directory hierarchy	`/` root, `/bin` binaries, `/etc` config, `/home` users, `/var` variable data, `/tmp` temp, `/opt` optional software
12	What does `df -h` show?	Disk filesystem usage (human-readable)
13	What does `du -sh /var/log` show?	Disk usage of specific directory
14	How to check which process has a file open?	`lsof /path/to/file`
15	What is `/etc/fstab`?	Filesystem mount table — defines auto-mounts at boot

Permissions (16-25)

#	Question	Answer
16	Explain `chmod 755`	Owner: rwx (7), Group: r-x (5), Others: r-x (5)
17	What is SUID?	SetUID — file runs as owner, not executor. `chmod u+s file`. Example: `/usr/bin/passwd`
18	What is SGID?	SetGID — file runs as group owner. On directory: new files inherit group
19	What is sticky bit?	On directory: only owner can delete files. Example: `/tmp` has sticky bit
20	Explain `umask 022`	Default permissions = 777 (dirs) or 666 (files) minus umask. 666-022 = 644 for files
21	`chown -R www-data:www-data /var/www`	Recursively change owner and group
22	What is ACL?	Access Control List — fine-grained permissions beyond owner/group/other. `setfacl`, `getfacl`
23	What does `chmod +x script.sh` do?	Adds execute permission for all (user, group, other)
24	Why can't a user delete a file in `/tmp` owned by another user?	Sticky bit is set on `/tmp`
25	How to find all SUID files?	`find / -perm -4000 -type f 2>/dev/null` — security audit command

Process Management (26-40)

#	Question	Answer / Command
26	List all processes	`ps aux` or `ps -ef`
27	Find process by name	`ps aux | grep nginx` or `pgrep -a nginx`
28	Kill a process gracefully	`kill <PID>` (sends SIGTERM — allows cleanup)
29	Force kill a process	`kill -9 <PID>` (SIGKILL — immediate, no cleanup)
30	What is a zombie process?	Process that has finished but parent hasn't collected exit status. Shows as `Z` in `ps`
31	How to fix zombie processes?	Kill the parent process. Init will adopt and reap orphan zombies
32	What is `nohup`?	Runs command immune to hangup signal. Persists after terminal close
33	Difference between `&` and `nohup`?	`&` backgrounds but dies with terminal. `nohup` survives terminal close
34	What does `nice` do?	Sets process priority. `-20` highest, `19` lowest. `nice -n 10 command`

#	Question	Answer / Command
35	What is `top` ? Real-time info shown?	Real-time process monitor: CPU%, MEM%, load average, uptime, process list
36	Explain load average (1, 5, 15 min)	Number of processes waiting for CPU. On 4-core: load 4.0 = 100% utilized
37	What is `htop` vs `top` ?	`htop` is interactive, colored, tree view, easier to use
38	How to run a process in background?	`command &` or `Ctrl+Z` then `bg`
39	What is `strace` ?	Traces system calls. Debug why a process fails: `strace -p <PID>`
40	What is `lsof` ?	List open files. Find what process uses a port: `lsof -i :8080`

Systemd & Services (41-50)

#	Question	Command
41	Start/stop/restart a service	`systemctl start/stop/restart nginx`
42	Enable service at boot	`systemctl enable nginx`
43	Check service status	`systemctl status nginx`
44	View service logs	`journalctl -u nginx -f` (follow)
45	View boot logs	`journalctl -b` (current boot)
46	List all running services	`systemctl list-units --type=service --state=running`
47	List failed services	`systemctl --failed`
48	Reload systemd after unit file change	`systemctl daemon-reload`
49	What is a systemd unit file?	Configuration defining how to manage a service. Located in `/etc/systemd/system/`
50	Create a custom systemd service	See unit file example below

```
# /etc/systemd/system/myapp.service
[Unit]
Description=My Application
After=network.target
Wants=postgresql.service

[Service]
Type=simple
User=appuser
WorkingDirectory=/opt/myapp
ExecStart=/opt/myapp/bin/server
Restart=on-failure
RestartSec=5
```

```
StandardOutput=journal
StandardError=journal
Environment=NODE_ENV=production

[Install]
WantedBy=multi-user.target
```

Networking (51-65)

#	Question	Command / Answer
51	Check IP address	`ip addr show` or `hostname -I`
52	Check open ports	`ss -tulnp` or `netstat -tulnp`
53	Check if port is reachable	`nc -zv host 443` or `curl -v telnet://host:443`
54	Check DNS resolution	`nslookup domain.com` or `dig domain.com`
55	Check routing table	`ip route show` or `route -n`
56	Trace packet path	`traceroute host` or `mtr host`
57	Check network interfaces	`ip link show`
58	What is `iptables`?	Linux firewall. Rules organized in chains: INPUT, OUTPUT, FORWARD
59	Block an IP with iptables	`iptables -A INPUT -s 192.168.1.100 -j DROP`
60	What is `firewalld`?	Dynamic firewall manager (frontend to nftables). `firewall-cmd --add-port=80/tcp`
61	Check TCP connections	`ss -tn` — shows established connections
62	What is `/etc/hosts`?	Local DNS override. Maps hostnames to IPs
63	What is `/etc/resolv.conf`?	DNS resolver configuration. `nameserver 8.8.8.8`
64	What's the difference between TCP and UDP?	TCP: reliable, ordered, connection-oriented. UDP: fast, no guarantee, connectionless
65	What is a network namespace?	Isolated network stack — used by Docker/K8s containers

Disk & Storage (66-75)

#	Question	Command
66	Check disk space	`df -h`
67	Check directory size	`du -sh /var/log/*`
68	Find largest files	`du -ah / | sort -rh | head -20`
69	List block devices	`lsblk`
70	Mount a disk	`mount /dev/sdb1 /mnt/data`
71	Create a filesystem	`mkfs.ext4 /dev/sdb1`

#	Question	Command
72	Check disk I/O	`iostat -x 1`
73	What is LVM?	Logical Volume Manager — flexible disk management. Resize volumes without unmount
74	Extend an LVM volume	`lvextend -L +10G /dev/vg/lv && resize2fs /dev/vg/lv`
75	What is swap?	Disk space used as virtual memory when RAM is full. `swapon --show`

Text Processing (76-85)

#	Command	Usage
76	`grep -r "error" /var/log/`	Recursively search for "error" in logs
77	`grep -i -c "error" file.log`	Case-insensitive count of "error" lines
78	`grep -E "error|warning" file.log`	Extended regex — match error OR warning
79	`sed 's/old/new/g' file`	Replace all occurrences of "old" with "new"
80	`sed -i '/^#/d' config.conf`	Delete comment lines in-place
81	`awk '{print $1, $4}' access.log`	Print 1st and 4th columns
82	`awk '/ERROR/ {count++} END {print count}' app.log`	Count ERROR lines
83	`cut -d: -f1 /etc/passwd`	Extract usernames from passwd
84	`sort file | uniq -c | sort -rn`	Count unique lines, sort by frequency
85	`tail -f /var/log/syslog | grep error`	Follow logs, filter for errors

SSH & Security (86-92)

#	Question	Answer
86	Generate SSH key pair	`ssh-keygen -t ed25519 -C "email@example.com"`
87	Copy SSH key to server	`ssh-copy-id user@host`
88	SSH tunneling	`ssh -L 8080:localhost:3000 user@host` — access remote 3000 via local 8080
89	What is SSH agent forwarding?	Allows using your local SSH keys on remote servers. `ssh -A user@host`
90	Disable root SSH login	Edit `/etc/ssh/sshd_config`: `PermitRootLogin no`
91	Harden SSH	Disable password auth, use key-only, change port, use fail2ban

#	Question	Answer
92	What is fail2ban?	Monitors logs and bans IPs with too many failed login attempts

Cron & Scheduling (93-96)

#	Question	Answer
93	Edit crontab	`crontab -e`
94	Cron syntax	`MIN HOUR DOM MON DOW command` — `*/5 * * * *` = every 5 min
95	Run daily at 2 AM	`0 2 * * * /opt/scripts/backup.sh`
96	Check cron logs	`journalctl -u cron` or `grep CRON /var/log/syslog`

Real Troubleshooting Scenarios (97-100)

#	Scenario	Diagnostic Commands	Resolution
97	Server slow, high CPU	`top` → identify process → `strace -p PID`	Kill runaway process, optimize code
98	Disk full, can't even run commands	`du -sh /var/log/*`, `lsof | grep deleted`	Delete old logs, truncate large files: `> /var/log/huge.log`
99	Can't SSH into server	`ping host`, `nc -zv host 22`, `journalctl -u sshd`	Check firewall, sshd status, disk space
100	High memory, system swapping	`free -h`, `ps aux --sort=-%mem | head`	Identify memory hog, increase RAM or optimize

---

Quick Command Cheat Sheet

```
# Performance
top / htop                    # Real-time process monitor
vmstat 1                      # Virtual memory stats every 1s
iostat -x 1                   # Disk I/O stats
free -h                       # Memory usage
uptime                        # Load average

# Files
find / -name "*.log" -mtime -1    # Files modified in last day
locate filename                    # Fast file search (uses db)
tar -czf archive.tar.gz /path     # Create compressed archive
tar -xzf archive.tar.gz           # Extract archive

# Network
curl -v https://api.example.com   # HTTP request with verbose
wget -q -O- http://localhost/health  # Quick health check
ss -tulnp                         # Open ports with process names
```

```
ip addr                          # Network interfaces
dig +short example.com           # DNS lookup
```